(a) 2 (b) $\sqrt{2}$ (c) 1 (d) $\frac{1}{\sqrt{2}}$

42. What is $\sqrt{15+\cot^2\left(\frac{\pi}{4}-2\cot^{-1}3\right)}$ equal to?

(a) 1 (b) 7 (c) 8 (d) 16

43. What is the value of sin 10° . sin 50° + sin 50° . sin 250° + sin 250° . sin 10° equal to?

(a) $-\frac{1}{4}$ (b) $-\frac{3}{4}$

(c) $\frac{3\sin 10°}{4}$ (d) $-\frac{3\cos 10°}{4}$

44. What is $\tan^{-1}\left(\frac{a}{b}\right)-\tan^{-1}\left(\frac{a-b}{a+b}\right)$ equal to?

(a) $-\frac{\pi}{4}$ (b) $\frac{\pi}{4}$

(c) $\tan^{-1}\left(\frac{a^2-b^2}{a^2+b^2}\right)$ (d) $\tan^{-1}\left(\frac{2ab}{a^2+b^2}\right)$

45. Under which one of the following conditions does the equation $(\cos\beta-1)x^2+(\cos\beta)x+\sin\beta=0$ in x have a real root for $\beta \in [0, \pi]$?

(a) $1-\cos\beta < 0$ (b) $1-\cos\beta \le 0$

(c) $1-\cos\beta > 0$ (d) $1-\cos\beta \ge 0$

46. In a triangle *ABC*, *AB* = 16 cm, *BC* = 63 cm and *AC* = 65 cm. What is the value of cos 2A + cos 2B + cos 2C?

(a) –1 (b) 0 (c) 1 (d) $\frac{76}{65}$

47. If $f(\theta) = \frac{1}{1+\tan\theta}$ and $\alpha+\beta = \frac{5\pi}{4}$, then what is the value of $f(\alpha)\,f(\beta)$?

(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$ (c) 1 (d) 2

48. If tan α and tan β are the roots of the equation $x^2-6x+8=0$, then what is the value of cos(2α + 2β)?

(a) $\frac{13}{75}$ (b) $\frac{13}{85}$ (c) $\frac{17}{85}$ (d) $\frac{19}{85}$

49. What is the value of tan 65° + 2tan 45° – 2tan 40° – tan 25°?

(a) 0 (b) 1 (c) 2 (d) 4

50. Consider the following statements:

1. In a triangle *ABC*, if cot *A* . cot *B* . cot *C* > 0, then the triangle is an acute angled triangle.
2. In a triangle *ABC*, if tan *A* . tan *B* . tan *C* > 0, then the triangle is an obtuse angled triangle.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

51. If (a, b) is the centre and c is the radius of the circle $x^2+y^2+2x+6y+1=0$, then what is the value of $a^2+b^2+c^2$?

(a) 19 (b) 18 (c) 17 (d) 11

52. If (1, –1, 2) and (2, 1, –1) are the end points of a diameter of a sphere $x^2+y^2+z^2+2ux+2vy+2wz-1=0$, then what is $u+v+w$ equal to?

(a) −2 (b) –1 (c) 1 (d) 2

53. The number of points represented by the equation $x = 5$ on the *xy*-plane is

(a) Zero (b) One

(c) Two (d) Infinitely many

54. If $< l, m, n >$ are the direction cosines of a normal to the plane $2x-3y+6z+4=0$, then what is the value of $49(7l^2+m^2-n^2)$?

(a) 0 (b) 1 (c) 3 (d) 71

55. A line through (1, –1, 2) with direction ratios < 3, 2, 2 > meets the plane $x+2y+3z=18$. What is the point of intersection of line and plane?

(a) (4, 4, 1) (b) (2, 4, 1)

(c) (4, 1, 4) (d) (3, 4, 7)

56. If p is the perpendicular distance from origin to the plane passing through (1, 0, 0), (0, 1, 0) and (0, 0, 1), then what is $3p^2$ equal to?

(a) 4 (b) 3 (c) 2 (d) 1

57. If the direction cosines $< l, m, n >$ of a line are connected by relation $l+2m+n=0$, $2l-2m+3n=0$, then what is the value of $l^2+m^2-n^2$?

(a) $\frac{1}{101}$ (b) $\frac{29}{101}$ (c) $\frac{41}{101}$ (d) $\frac{92}{101}$

58. If a variable line passes through the point of intersection of the lines $x+2y-1=0$ and $2x-y-1=0$ and meets the coordinate axis in *A* and *B*, then what is the locus of the mid-point of *AB*?

(a) $3x+y=10xy$ (b) $x+3y=10xy$

(c) $3x+y=10$ (d) $x+3y=10$

59. What is the equation to the straight line passing through the point (–sinθ, cosθ) and perpendicular to the line $x\cos\theta + y\sin\theta = 9$?

(a) $x\sin\theta - y\cos\theta - 1 = 0$

(b) $x\sin\theta - y\cos\theta + 1 = 0$

(c) $x\sin\theta - y\cos\theta = 0$

(d) $x\cos\theta - y\sin\theta + 1 = 0$